

MID TERM EXAMINATION

Course: Data Structures (CS101)

Set: A | Time: 3 Hours | Max Marks: 128 | Date: 15-03-2026

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

GENERAL INSTRUCTIONS:
1. Answer all questions.
2. Figures to the right indicate full marks.

SECTION A: MCQs

1. Which of the following is NOT explicitly mentioned as a type of scenario for architectural evaluation? [3 Marks]

- a) Use-case scenarios
- b) Growth scenarios
- c) Performance scenarios
- d) Exploratory scenarios

2. Which factor directly implies that the system design must adapt to the component's existing functionality? [3 Marks]

- a) Faster development
- b) Limited customization
- c) Architecture must adjust to the component
- d) Vendor dependency

3. In architectural design decision-making, which of the following is an example of a Non-Functional Requirement (Quality Attribute)? [3 Marks]

- a) Inventory management
- b) Product search
- c) Scalability
- d) Payments

SECTION B: SHORT ANSWERS

4. For a successful evaluation of architecture, what must be clearly defined and measurable? Provide an example from

the text.

[6 Marks]

5. List three essential items that complete, accurate, and up-to-date architecture documentation should include.**[6 Marks]**
6. Explain what "Extra integration work" entails as an impact of using off-the-shelf components. **[6 Marks]**
7. What does "Vendor dependency" imply when an off-the-shelf component is used in an architecture? **[6 Marks]**
8. According to the text, what are "Risks" and "Sensitivity points" in the context of architectural evaluation? **[6 Marks]**
9. Name three key stakeholders whose involvement is required for successful architectural evaluation. **[6 Marks]**
10. Briefly describe what "Reusability and Extensibility" refers to as a factor in design decision-making. **[6 Marks]**

SECTION C: LONG ANSWERS

11. Discuss the three types of scenarios important for architectural evaluation and explain the purpose of each**[11 Marks]**
 12. Explain the importance of an "Experienced Evaluation Team" in the successful evaluation of architecture. What specific areas of understanding should such a team possess? **[11 Marks]**
 13. Analyze the implications of "Limited customization" and "Hard to replace later" as impacts of using off-the-shelf components in software architecture. **[11 Marks]**
 14. Discuss how using off-the-shelf components can lead to both "Faster development" and "Extra integration work," providing an example for each from the text. **[11 Marks]**
 15. Elaborate on the distinct roles of "Functional Requirements" and "Non-Functional Requirements" (Quality Attributes) in guiding design decisions in software architecture. **[11 Marks]**
 16. Discuss how "Budget and Schedule" and "Team Skills and Experience" influence architectural design decisions, providing specific examples from the text for each. **[11 Marks]**
 17. Compare and contrast "Structured Evaluation Methods" and "Proper Tools and Techniques" as factors for the successful evaluation of architecture. **[11 Marks]**
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